

Comparing All Seven Parent Functions

TEACHER KEY | Algebra II | Unit 1 | Day 8 | TEK A2.2.A

Objective: I can compare the key attributes of all seven parent functions, name a function from its attributes alone, and find a relative maximum or minimum on a given interval.

Vocabulary

Relative maximum — the **largest** y-value on a given interval.

Relative minimum — the **smallest** y-value on a given interval.

A function can have no overall max or min but still have one **on an interval**.

Part 1 — The Master Comparison Table

Fill this in from your Day 4-7 notes. This is your reference sheet for the rest of the year.

Function	Domain	Range	Symmetry	Asymptotes
$\sqrt{x}$	$[0, \infty)$	$[0, \infty)$	none	none
$ x $	$(-\infty, \infty)$	$[0, \infty)$	y-axis	none
x^3	$(-\infty, \infty)$	$(-\infty, \infty)$	rotational	none
$\sqrt[3]{x}$	$(-\infty, \infty)$	$(-\infty, \infty)$	rotational	none
$1/x$	$x \neq 0$	$y \neq 0$	both kinds	$x = 0, y = 0$
b^x	$(-\infty, \infty)$	$(0, \infty)$	none	$y = 0$
log base b	$(0, \infty)$	$(-\infty, \infty)$	none	$x = 0$

Part 2 — Intercepts and Extremes

Function	Intercept(s)	Minimum?	Maximum?
$\sqrt{x}$	$(0, 0)$	yes, $y = 0$	no
$ x $	$(0, 0)$	yes, $y = 0$	no
x^3 and $\sqrt[3]{x}$	$(0, 0)$	no	no
$1/x$	none	no	no
b^x	$(0, 1)$	no	no
log base b	$(1, 0)$	no	no

Only **two** of the seven have a minimum, and **none** has a maximum. Only **$1/x$** has no intercept at all.

Part 3 — Name That Function

Use only the attributes listed. Name the function and sketch its shape in the box.

- Domain and range both all reals except 0; two branches; asymptotes at $x = 0$ and $y = 0$.

Function: **$1/x$**

- Domain all reals, range $[0, \infty)$, folds across the y-axis, minimum of 0.

Function: **$|x|$**

- Domain $(0, \infty)$, range all reals, passes through $(1, 0)$.

Function: **log base b**

- Domain and range both $[0, \infty)$, no symmetry, no asymptotes.

Function: $\sqrt{x}$

Part 4 — Relative Max and Min on an Interval

A graph rises and falls between $y = 1.5$ and $y = -1.5$ over the interval from 0° to 360° .

Relative maximum on that interval: $y = 1.5$ Relative minimum: $y = -1.5$

Now consider $f(x) = x^3$ on the interval $[-2, 2]$.

Relative maximum: $y = 8$, at $x = 2$ Relative minimum: $y = -8$, at $x = -2$

x^3 has no overall max or min, but on a **closed interval** it has both. Why? **the endpoints limit it**

Part 5 — Sort by One Attribute

Which parent functions have a restricted domain? $\sqrt{x}$, $1/x$, log base b

Which have any asymptote? $1/x$, b^x , log base b

Which have range all real numbers? x^3 , $\sqrt[3]{x}$, log base b

Practice

Use your comparison tables. Show reasoning where there is room.

- Two parent functions pass through $(1, 1)$ and $(0, 0)$. Name them. $\sqrt{x}$, $|x|$, x^3 , $\sqrt[3]{x}$
- Which parent function never touches either axis? $1/x$
- Which two are reflections of each other across $y = x$? Give two different pairs.
 x^3 and $\sqrt[3]{x}$; also b^x and log base b
- A graph has a minimum but no maximum, and no asymptotes. Name both possibilities.
 $\sqrt{x}$ and $|x|$
- $f(x) = |x|$ on the interval $[-3, 1]$: minimum **0** maximum **3**
- Explain why a function can have a relative maximum on an interval but none overall.

Exit Ticket

Two parent functions have domain and range that are both all real numbers, and both have 180° rotational symmetry.

Name them: x^3 and $\sqrt[3]{x}$ Name one function with a restricted range: $\sqrt{x}$, $|x|$, or b^x